



Web Designing -I

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Unit 1. Basic Web Technology

Topic 01: Internet Basic Concepts :

Internet is defined as an Information super Highway, to access information over the web.

However, It can be defined in many ways as follows:

- ☐ Internet is a world-wide global system of interconnected computer networks.
- ☐ Internet uses the standard Internet Protocol (TCP/IP).
- ☐ Every computer in internet is identified by a unique IP address.



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Internet Basic Concepts :

- ❑ IP Address is a unique set of numbers (such as 110.22.33.114) which identifies a computer location
- ❑ A special computer DNS (Domain Name Server) is used to give name to the IP Address so that user can locate a computer by a name.
- ❑ For example, a DNS server will resolve a name <http://www.amazon.com> to a particular IP address to uniquely identify the computer on which this website is hosted.

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Internet is accessible to every user all over the world.





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Evolution of Internet :

- ☐ The concept of Internet was originated in 1969 and has undergone several technological & Infrastructural changes as discussed below:
- ☐ The origin of Internet devised from the concept of **Advanced Research Project Agency Network (ARPANET)**.
- ☐ **ARPANET** was developed by United States Department of Defense.
- ☐ Basic purpose of ARPANET was to provide communication among the various bodies of government.



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Evolution of Internet :

- ❑ Initially, there were only four nodes, formally called **Hosts**.
- ❑ In 1972, the **ARPANET** spread over the globe with 23 nodes located at different countries and thus became known as **Internet**.
- ❑ By the time, with invention of new technologies such as TCP/IP protocols, DNS, WWW, browsers, scripting languages etc. Internet provided a medium to publish and access information over the web.



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Internet Services:

- Internet Banking
- Matrimonial Services
- Online Shopping
- Online Ticket Booking
- Online Bill Payment
- Data Sharing
- E-mail



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Topic 02: URL: (Uniform Resource Locator)

Every document on the Web has a unique address. This address is known as Uniform Resource Locator (URL).

A URL is made of up several parts, each of which offers information to the web browser to help find the page. It is easier to learn the parts of a URL, if you look at the example URL given below, there are three key parts: the **scheme**, the **host address**, and the **file path**.

`https://mail.google.com/mail`



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URL: (Uniform Resource Locator)

1. Scheme:

The scheme identifies the type of protocol and URL you are linking to and therefore, how the resource should be retrieved.

http://

Hypertext Transfer Protocol (HTTP) is used to request pages from Web servers and send them back from Web servers to browsers.

https://

Secure Hypertext Transfer Protocol (HTTPS) encrypts the data sent between the browser and the Web server using a digital certificate.



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URL: (Uniform Resource Locator)

1. Scheme:

ftp://

File Transfer Protocol is another method for transferring files on the Web. While HTTP is a lot more popular for viewing Web sites because of its integration with browsers, FTP is still commonly used protocol to transfer large files across the Web and to upload source files to your Web server.



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URL: (Uniform Resource Locator)

2. The Host Address

The host address is where a website can be found, either the IP address (four sets of numbers between 0 and 255, for example 68.178.157.132) or more commonly the domain name for a site such as `www.tutorialspoint.com`. Note that "www" is not actually part of the domain name although it is often used in the host address



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URL: (Uniform Resource Locator)

3. The File Path

The file path always begins with a forward slash character, and may consist of one or more directory or folder names. Each directory name is separated by forward slash characters and the file path may end with a filename at the end. Here index.htm is the filename which is available in html directory:

`https://www.tutorialspoint.com/html/index.htm`



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Topic 03: ISP: (Internet Service Provider)

With the [Internet](#) becoming one of the vital services of our daily routine, the Internet bill has joined the set of compulsory monthly expenses, including the electricity bill and the various other taxes. To pay our Internet bill we turn to our **ISP, the company that is responsible for delivering Internet access to our place.**

The ISP, short for 'Internet Service Provider', is the company that connects your personal computer, notebook, tab, PDA, mobile device, game station, etc. to the Internet. Another name **for ISP is IAP, or Internet Access Provider.**



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ISP: (Internet Service Provider)

An Internet Service Provider (ISP) is a company that provides Internet access by using **copper, fiber, or even satellite communications to the customer**. In this lesson, we will cover exactly what an ISP is, how the ISP connects customers, and what type of services the ISPs provide.

An **Internet Service Provider (ISP)** is a company such as AT&T, Verizon, Comcast, or Bright House, Idea, that provides Internet access to companies, families, and even mobile users. ISPs use fiber-optics, satellite, copper wire, and other forms to provide Internet access to its customers.



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ISP: (Internet Service Provider)

The connection between your Internet enabled device and the global network is executed through a specific **digital data transmission technology**. It represents the transfer of **information packets** through an Internet **Protocol route**.

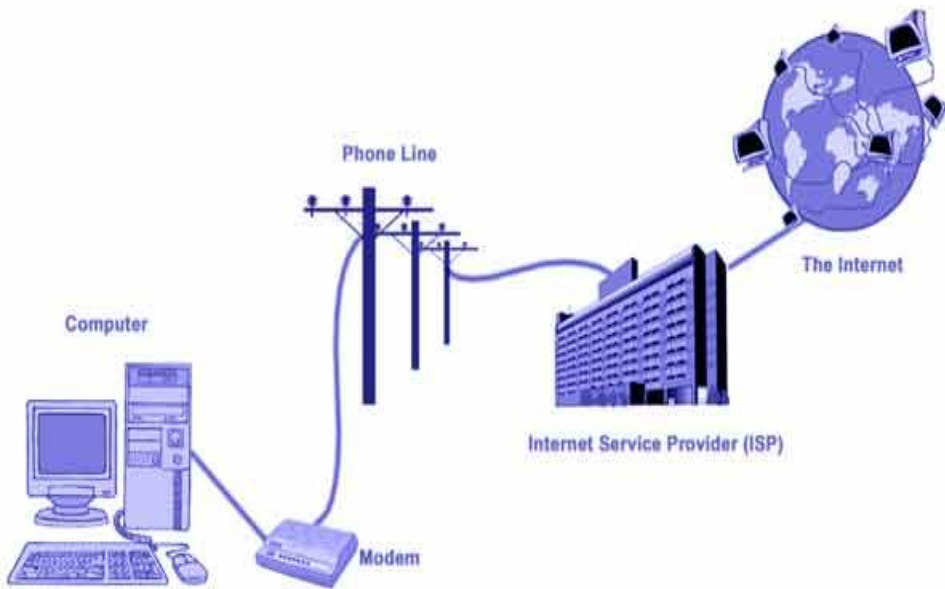
According to the method of data transmission, the Internet access that ISPs provide to users can be divided into several types, the most popular of which are:





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ISP: (Internet Service Provider)



Dial-up Internet access

This is the oldest method of providing access to the Internet. It uses a **telephone line to perform a modem-to-modem** connection. For that purpose, the user's computer is attached to a telephone line enabled modem device, which dials into the node of the ISP and starts transferring data between the servers that store websites the user wants to see and their Internet connected device.

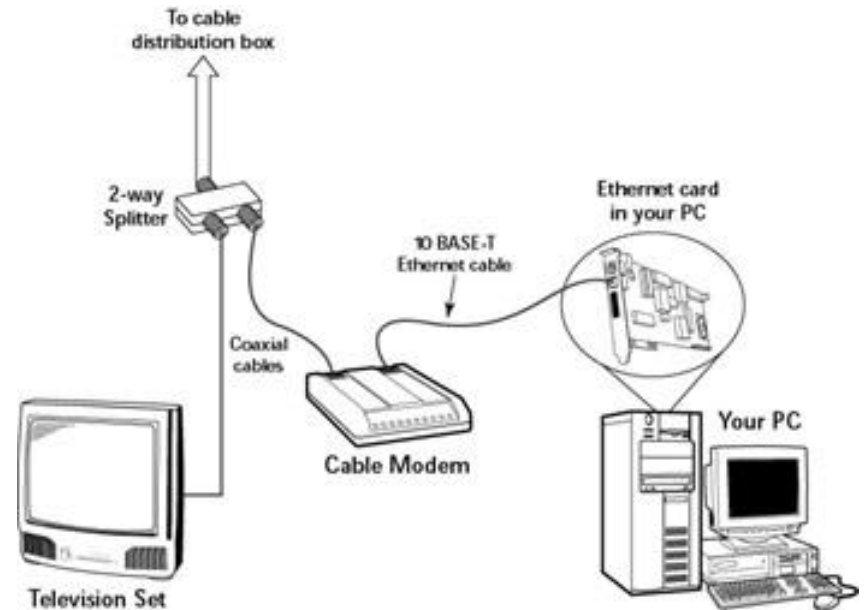


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ISP: (Internet Service Provider)

Cable Internet

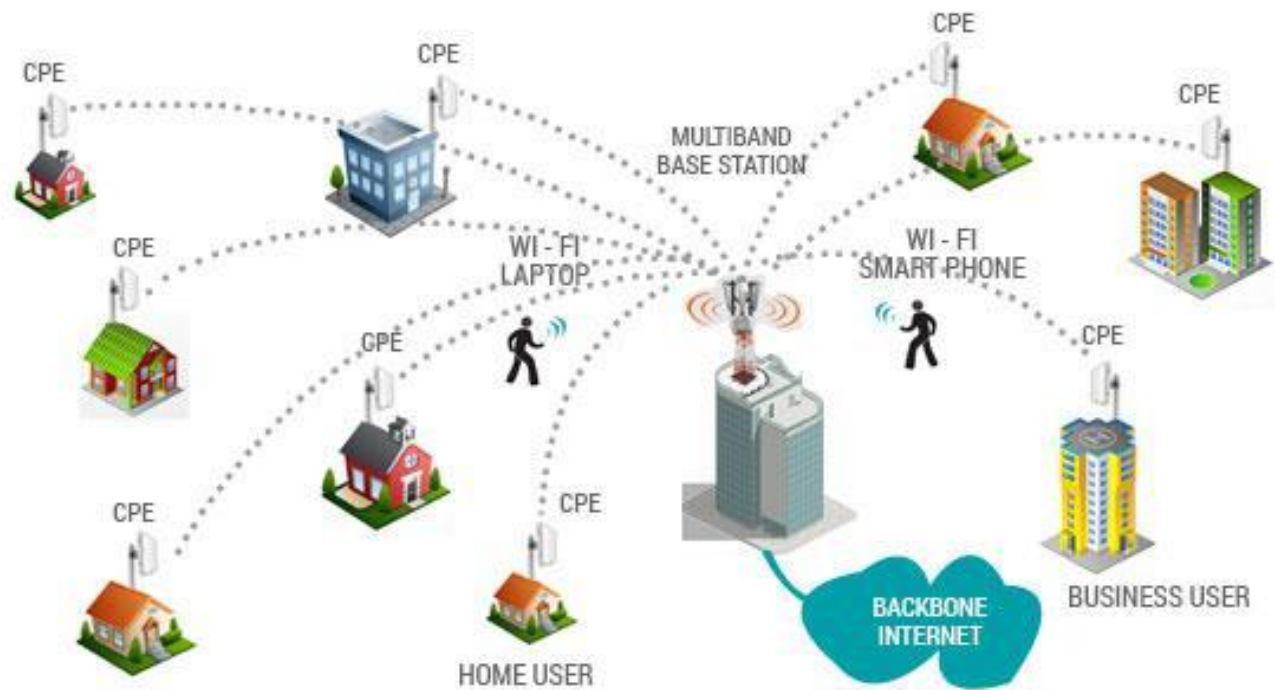
The cable Internet is among the most preferred methods for providing residential Internet access. Technically speaking, it represents a broadband Internet access method, using the high-bandwidth cable television network to transmit data between the global network and the households.





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ISP: (Internet Service Provider)





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ISP: (Internet Service Provider)

Wi-Fi Internet

Wi-Fi (from Wireless Fidelity) has become one of the most widely distributed Internet access methods, with the growing usage of **portable computers and Internet enabled mobile devices, such as smart phones, PDAs, game consoles, etc.** In this sense, it is the most mobile Internet access method, since you are able to use it everywhere as long as you are located within the scope of coverage, i.e. within the range of an Internet connected wireless network. Due to its ability to serve mobile devices, **Wi-Fi is used in public places such as airports, hotels and restaurants to provide Internet access to customers.** There are also specialized Wi-Fi hotspots where the service is either free or paid. **Some of the largest cities in the world are in the process of building Wi-Fi networks that cover all the public places in the central areas.**



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Topic 04: Intranet & Extranet :

An intranet is a private network that allows employees and staff in an enterprise to securely share knowledge and information easily within the company or organization. Information, tools, directories, and services available on a company's intranet are typically unavailable to the general public. The prefix "intra" implies that an intranet is designed for **internal communications only**.

An extranet is a private network that leverages internet technology and public telecommunication system to share part of a business's information or operations over a secure system with suppliers, vendors, partners, customers, or other businesses. **An extranet is often considered part of a company's intranet that is extended to authorized users outside of the organization.**



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Intranet :

Communicate within the business: both top-down and bottom-up by distributing news and announcements, providing feedback and sharing information

Manage documents and business information: intranets offer a centralized location to host, search and access vital business content

Enable collaboration: functionality such as team areas, forums, blogs and social tools can support employees to work together effectively, regardless of their department or location

Support strategic business objectives: these may span improved [employee engagement](#) and morale, embedding a business culture, increasing staff retention or encouraging better productivity and efficiency



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Extranet :

extranets take the concept of an intranet to another level, **allowing customers, vendors, partners or selected third parties access to a company intranet, or selected areas of it.**

This can prove extremely valuable in organizations where you need to share information or confidential data with a third party, by providing a more secure and efficient process than traditional channels such as email.

For example, an organization can provide extranet access for a supplier to manage ordering, tracking and inventory management, improving efficiency and lowering business overheads. In a collaborative industry such as healthcare, extranets can be used to bring together different care providers or Trusts to share best practice and improve business growth.



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Difference between Internet , Intranet & Extranet :

Internet	Intranet	Extranet
It is a global system of interconnected computer network.	It is a private network specific to an organization	It is a private network that use public network to share information with suppliers and vendors.
No regulated by any authority.	It is regulated by an organization	It is regulated by multiple organization
Thus content in the network accessible to everyone connected.	Thus content in the network accessible only to members of organization.	Thus content in the network accessible to members of organization & External members.



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Internet	Intranet	Extranet
It is largest in terms of number of connected devices.	It is small network with minimal number of connected devices.	It is bigger network compare to intranet.
It is Owned by no one.	It is Owned by single organization.	It is owned by single or multiple organization.
It is means of sharing information throughout the world.	It is means of sharing sensitive information within organization.	It is means of sharing information within organization and external members.



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Internet	Intranet	Extranet
Security is dependent on number of connected networked.	Security is enforced via a firewall	Security is enforced via a firewall that separates internet and Extranet
What we are normally using is internet.	TCS using internal networks of its business operation.	HP and Intel Using network for business related operation.
Internet is unregulated.	Intranet is regulated by organization policies.	Extranet is regulated by contractual agreement between organization.



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Topic 05: Hypertext Transfer Protocol:

The Hypertext Transfer Protocol (HTTP) is an application-level protocol for distributed, collaborative, hypermedia information systems. **This is the foundation for data communication for the World Wide Web (i.e. internet) since 1990.** HTTP is a generic and stateless protocol which can be used for other purposes as well using extensions of its request methods, error codes, and headers.

Basically, HTTP is a TCP/IP based communication protocol, that is used to deliver data (HTML files, image files, query results, etc.) on the World Wide Web. The default port is TCP 80, but other ports can be used as well. It provides a standardized way for computers to communicate with each other. HTTP specification specifies how clients' request data will be constructed and sent to the server, and how the servers respond to these requests.



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Hypertext Transfer Protocol: PORT 80

Port 80 is the port number assigned to commonly used internet communication protocol, Hypertext Transfer Protocol (HTTP). It is the port from which a computer sends and receives Web client-based communication and messages from a Web server and is used to send and receive HTML pages or data.

Port 80 is one of the most commonly used port numbers in the Transmission Control Protocol (TCP) suite. Any Web/HTTP client, such as a Web browser, uses port 80 to send and receive requested Web pages from a HTTP server. It manages all HTTP-based requests that originate from a computer, regardless of the number of requests and initiating Web clients. Similarly, the HTTP server responds to all requests received at port 80.



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Features of Hypertext Transfer Protocol:

There are three basic features that make HTTP a simple but powerful protocol:

HTTP is connectionless: The HTTP client, i.e., a browser initiates an HTTP request and after a request is made, the client disconnects from the server and waits for a response. The server processes the request and re-establishes the connection with the client to send a response back.

HTTP is media independent: It means, any type of data can be sent by HTTP as long as both the client and the server know how to handle the data content. It is required for the client as well as the server to specify the content type using appropriate MIME-type.



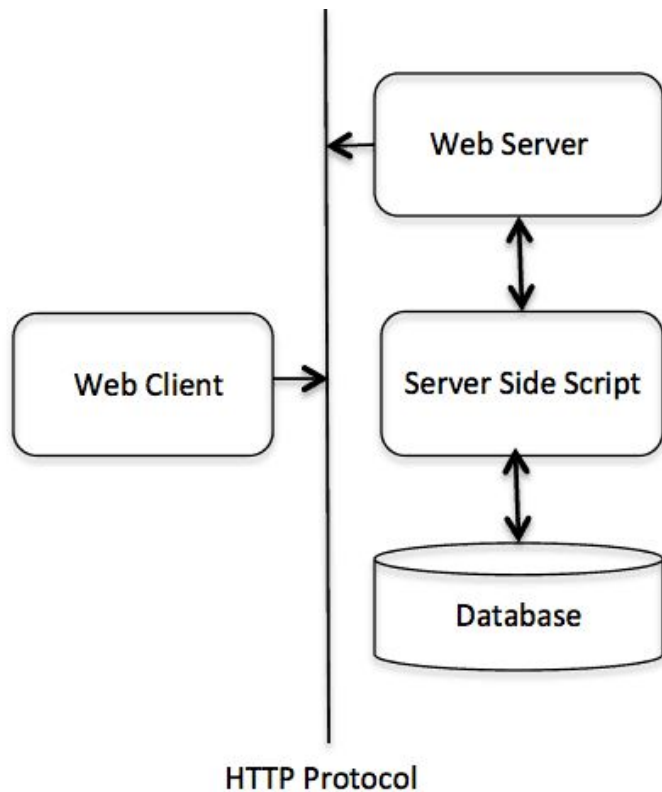
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Features of Hypertext Transfer Protocol:

HTTP is stateless: As mentioned above, HTTP is connectionless and it is a direct result of HTTP being a stateless protocol. The server and client are aware of each other only during a current request. Afterwards, both of them forget about each other. Due to this nature of the protocol, neither the client nor the browser can retain information between different requests across the web pages.



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The HTTP protocol is a request/response protocol based on the client/server based architecture where web browsers, robots and search engines, etc. act like HTTP clients, and the Web server acts as a server.



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Client

The HTTP client sends a request to the server in the form of a request method, URI, and protocol version, followed by a MIME (Multi-Purpose Internet Mail Extensions) like message containing request modifiers, client information, and possible body content over a TCP/IP connection.

Server

The HTTP server responds with a status line, including the message's protocol version and a success or error code, followed by a MIME-like message containing server information, entity meta information, and possible entity-body content.



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Topic 06: World Wide Web:

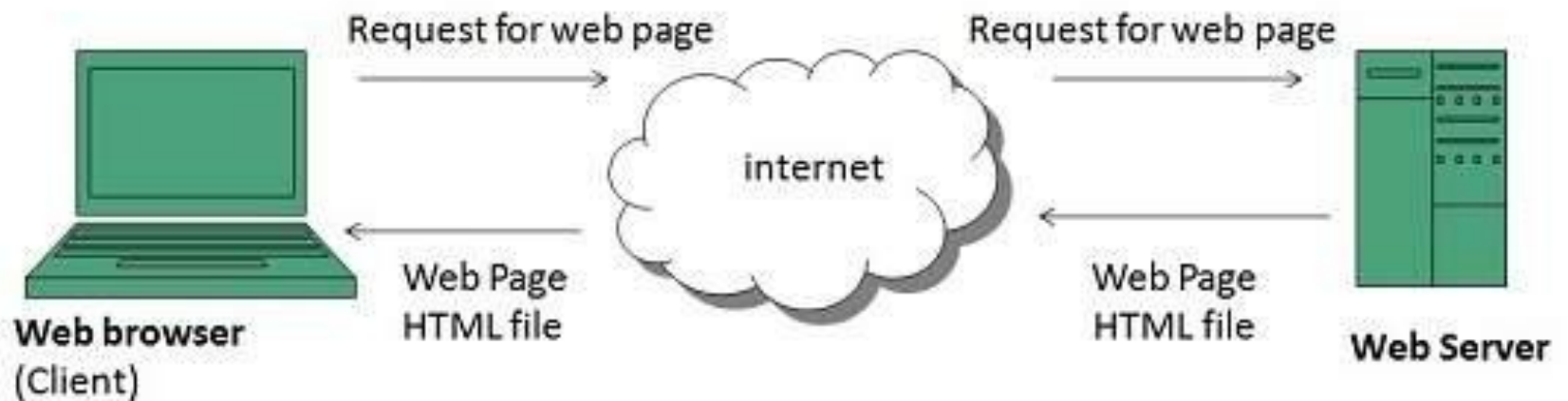
WWW stands for **World Wide Web**. A technical definition of the World Wide Web is : all the resources and users on the Internet that are using the Hypertext Transfer Protocol (HTTP).

The World Wide Web is a way of exchanging information between computers on the Internet, tying them together into a vast collection of interactive multimedia resources.



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World Wide Web:





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World Wide Web:

WWW works on client- server approach. Following steps explains how the web works:
User enters the URL (say, **<http://www.dcs.gnu.ac.in>**) of the web page in the address bar of web browser.

Then browser requests the **Domain Name Server** for the IP address corresponding to **www.dcs.gnu.ac.in**

After receiving IP address, browser sends the request for web page to the web server using HTTP protocol which specifies the way the browser and web server communicates.

Then web server receives request using HTTP protocol and checks its search for the requested web page. If found it returns it back to the web browser and close the HTTP connection.

Now the web browser receives the web page, It interprets it and display the contents of web page in web browser's window.



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Topic 07: Search Engine:

Search Engine refers to a huge database of internet resources such as web pages, newsgroups, programs, images etc. It helps to locate information on World Wide Web.

User can search for any information by passing query in form of keywords or phrase. It then searches for relevant information in its database and return to the user.



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Search Engine Component:

Generally there are three basic components of a search engine as listed below:

1. Web Crawler
2. Database
3. Search Interfaces

Web crawler

It is also known as **spider** or **bots**. It is a software component that traverses the web to gather information.

Database

All the information on the web is stored in database. It consists of huge web resources.

Search Interfaces

This component is an interface between user and the database. It helps the user to search through the database.



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Search Engine Working:

Web crawler, database and the search interface are the major component of a search engine that actually makes search engine to work. Search engines make use of Boolean expression AND, OR, NOT to restrict and widen the results of a search. Following are the steps that are performed by the search engine:

The search engine looks for the keyword in the index for predefined database instead of going directly to the web to search for the keyword.

It then uses software to search for the information in the database. This software component is known as web crawler.

Once web crawler finds the pages, the search engine then shows the relevant web pages as a result. These retrieved web pages generally include title of page, size of text portion, first several sentences etc.



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Search Engine Working:

Search Engine	Description
Google	It was originally called BackRub . It is the most popular search engine globally.
Bing	It was launched in 2009 by Microsoft . It is the latest web-based search engine that also delivers Yahoo's results.
Ask	It was launched in 1996 and was originally known as Ask Jeeves . It includes support for match, dictionary, and conversation question.
AltaVista	It was launched by Digital Equipment Corporation in 1995. Since 2003, it is powered by Yahoo technology.
Alexa	It is subsidiary of Amazon and used for providing website traffic information.



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Topic 08: Web Server & Web Browser:





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Web Server:

Web server is a computer where the web content is stored. Basically web server is used to host the web sites but there exists other web servers also such as gaming, storage, FTP, email etc.

Web site is collection of web pages while web server is a software that respond to the request for web resources.

Web server respond to the client request in either of the following two ways:

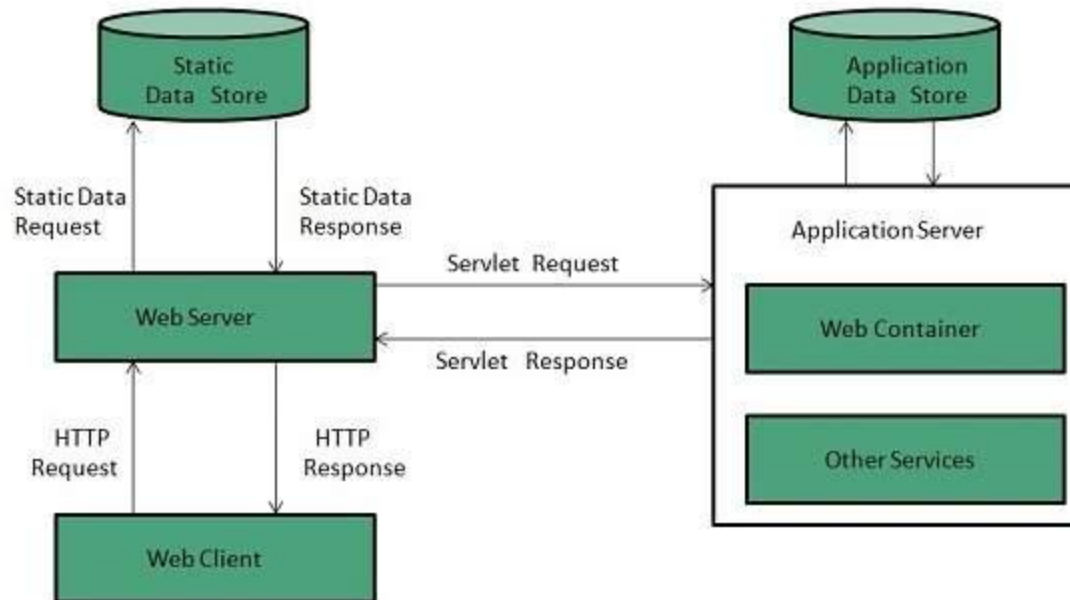
Sending the file to the client associated with the requested URL.

Generating response by invoking a script and communicating with database



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Web Browser & Web Server:





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Web Browser & Web Server:

When client sends request for a web page, the web server search for the requested page if requested page is found then it will send it to client with an HTTP response.

If the requested web page is not found, web server will the send an **HTTP response: Error 404 Not found**.

If client has requested for some other resources then the web server will contact to the application server and data store to construct the HTTP response.



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Web Server Architecture:

Web Server Architecture follows the following two approaches:

1. Concurrent Approach
2. Single-Process-Event-Driven Approach.

Concurrent approach allows the web server to handle multiple client requests at the same time. It can be achieved by following methods:

Multi-process
Multi-threaded
Hybrid method.



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Web Server Examples:

Apache HTTP Server:

This is the most popular web server in the world developed by the Apache Software Foundation. Apache web server is an open source software and can be installed on almost all operating systems including Linux, UNIX, Windows, FreeBSD, Mac OS X and more. About 60% of the web server machines run the Apache Web Server.

Sun Java System Web Server

This web server from Sun Microsystems is suited for medium and large web sites. Though the server is free it is not open source. It however, runs on Windows, Linux and UNIX platforms. The Sun Java System web server supports various languages, scripts and technologies required for Web 2.0 such as JSP, Java Servlets, PHP, Perl, Python, and Ruby on Rails, ASP and Coldfusion etc.



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Web Browser:

web Browser is an application software that allows us to view and explore information on the web. User can request for any web page by just entering a URL into address bar.

Web browser can show text, audio, video, animation and more. It is the responsibility of a web browser to interpret text and commands contained in the web page.

Earlier the web browsers were text-based while now a days graphical-based or voice-based web browsers are also available. Following are the most common web browser available today:



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Web Browser:

Browser	Vendor
Internet Explorer	Microsoft
Google Chrome	Google
Mozilla Firefox	Mozilla
Netscape Navigator	Netscape Communications Corp.
Opera	Opera Software
Safari	Apple
Sea Monkey	Mozilla Foundation
K-meleon	K-meleon



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Web Browser Architecture:

There are a lot of web browser available in the market. All of them interpret and display information on the screen however their capabilities and structure varies depending upon implementation. But the most basic component that all web browser must exhibit are listed below:

1. Controller/Dispatcher
2. Interpreter
3. Client Programs



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Web Browser Architecture:

Controller works as a control unit in CPU. It takes input from the keyboard or mouse, interpret it and make other services to work on the basis of input it receives.

Interpreter receives the information from the controller and execute the instruction line by line. Some interpreter are mandatory while some are optional For example, HTML interpreter program is mandatory and java interpreter is optional.

Client Program describes the specific protocol that will be used to access a particular service. Following are the client programs tat are commonly used:

HTTP

SMTP

FTP

NNTP

POP



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Topic 09: Web Pages: Static & Dynamic Web Pages.

Website:

Website is a collection of related web pages that may contain text, images, audio and video. The first page of a website is called home page. Each website has specific internet address (URL) that you need to enter in your browser to access a website.

Website is hosted on one or more servers and can be accessed by visiting its homepage using a computer network. A website is managed by its owner that can be an individual, company or an organization.

Collection of linked web pages on a web server is known as **website**. There is unique **Uniform Resource Locator (URL)** is associated with each web page.



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Web Pages: Static & Dynamic Web Pages.

A website can be of two types:

1. Static Website
2. Dynamic Website



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Static Website:

Static website is the basic type of website that is easy to create. You don't need the knowledge of web programming and database design to create a static website. Its web pages are coded in HTML.

The codes are fixed for each page so the information contained in the page does not change and it looks like a printed page.



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Dynamic Website.

Dynamic website is a collection of dynamic web pages whose content changes dynamically. It accesses content from a database or Content Management System (CMS). Therefore, when you alter or update the content of the database, the content of the website is also altered or updated.

Dynamic website uses client-side scripting or server-side scripting, or both to generate dynamic content.

Client side scripting generates content at the client computer on the basis of user input. The web browser downloads the web page from the server and processes the code within the page to render information to the user.

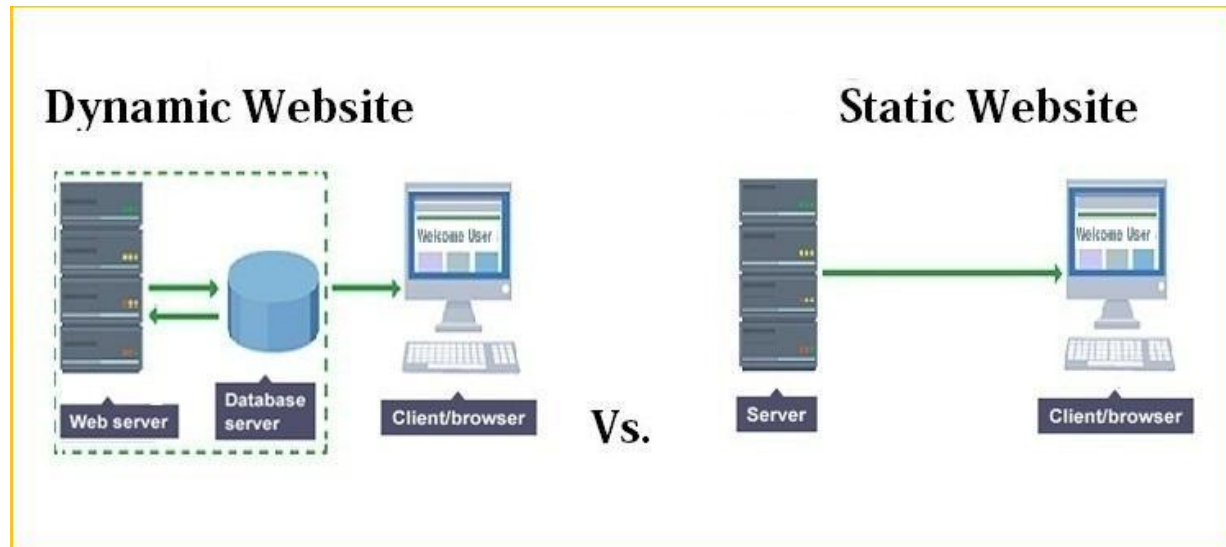
In server side scripting, the software runs on the server and processing is completed in the server then plain pages are sent to the user.



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Static & Dynamic Website.

You can often tell if a page is static or dynamic simply by looking at the page's [file extension](#) in the [URL](#), located in the address field of the Web browser. If it is ".htm" or ".html," the page is probably static. If the extension is ".php," ".asp," or ".jsp," the page is most likely dynamic.





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Difference Between Static & Dynamic Website.

- In static web pages the theme and content of web pages remained fixed and in dynamic web pages they changed according to run time.
- Browsing and loading of static web pages are more quicker than dynamic web pages.
- Changing the content in static web pages is a difficult task because you have to develop and upload a new page while in dynamic web pages server application do this automatically.



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Difference Between Static & Dynamic Website.

- If file extension of a URL is in .htm or .html then it is a static web pages. While if it is in .php, .asp and .jsp then it is the example of dynamic web pages.
- Static web pages are created through HTML language while dynamic web pages are created by the usage of PHP, JavaScript and Action script languages.
- Static web pages plan is an easy and cheap method if you wants to create a static and non updating web pages. While dynamic web pages method is advisable if you have plan to update the content and material frequently.



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Topic 10: Types of Website:

- Blog or Personal Website
- Business Website
- E -Commerce
- Business Directory
- Question & Answer Website / Informational
- Non Profits and Religious Websites
- Online Communities
- Forum



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Types of Website:

Internet Forum:

An internet forum is message board where people can hold conversation by posting messages.

- A forum can contain several sub forums.
- Each of sub forums may contain a number of topics.
- Within a forum's topic, each new discussion started is called a thread.
- This thread can be replied by as many people as so wish.





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Types of Website:

Blog:

The term **Blog** is taken from **web log**. It is a kind of web site that is updated regularly, with content about almost anything. In other words, blog is a **Content Management System (CMS)**, an easy way of publishing articles on the internet.

- Write what you know about. For example, if you have good computer knowledge. You can write what you know about the subject.
- You can share your experience. You can also write what you gained from that experience, what you learned.
- Detail your personal research.
- Share your memory of someone.





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Types of Website:

E-commerce:

E-Commerce or **Electronics Commerce** is a methodology of modern business which addresses the need of business organizations, vendors and customers to reduce cost and improve the quality of goods and services while increasing the speed of delivery. E-commerce refers to paperless exchange of business information using following ways.

Electronic Data Exchange (EDI)

Electronic Mail (e-mail)

Electronic Bulletin Boards

Electronic Fund Transfer (EFT)

Other Network-based technologies



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Types of Website:

Online Business Brochure / Catalog :

A brochure website is **the simplest type of website** in terms of functionality. Brochure websites typically only have a **few pages**, and will be used by **small businesses** that need a **simple online presence**. For example, a small plumbing company would only need a brochure website with a homepage displaying contact information, an 'about us' page and perhaps a couple of photos of their work. Their website is like an **online business card** for potential customers.





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Types of Website:

Portal:

A portal website **brings together information from lots of different sources** on the web. Early examples include **Aol** and **Yahoo**, who offer emails, forums, search engines and news all through their homepage. Portals can also be for internal use in a school, university or company, allowing students or employees to access their emails, alerts and files all in one place.





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Types of Website:

Wiki / Informational Website:

A wiki website is one which **allows people to collaborate online and write content together**. The most popular example is Wikipedia itself, which allows anyone to amend, add and assess the content of their articles.



WIKIPEDIA
The Free Encyclopedia



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Types of Website:

Social media:

Social media websites are platforms which allow the **sharing of images or ideas**. They encourage **online interaction and sharing**. The most popular social media website is **Facebook**, with a staggering **2.07 billion active users**. Other social media sites include **YouTube**, **Twitter**, **Instagram** and **LinkedIn**.





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Types of Website:

Photo Sharing Websites:

There are web companies like, [Flickr.com](https://www.flickr.com), [Photosite.com](https://www.photosite.com), and Google's [Picasa](https://picasa.google.com). There could easily be over a hundred such sites that offer free photo sharing paid for by their online advertising. Also, many digital cameras and photo printers now come with software enabling mere mortals to create digital photo slide shows and upload them to the web.





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Topic 11. Web Designing Principles :

Websites that are not well designed tend to perform poorly and have sub-optimal Google Analytics metrics (e.g. high bounce rates, low time on site, low pages per visit and low conversions). So what makes good web design?

Below we explore the top 10 web design principles that will make your website aesthetically pleasing, easy to use, engaging, and effective.



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Web Designing Principles :

1. PURPOSE

Good web design always caters to the needs of the user. Are your web visitors looking for information, entertainment, some type of interaction, or to transact with your business? Each page of your website needs to have a clear purpose, and to fulfill a specific need for your website users in the most effective way possible.

DESIGN

IS NOT JUST WHAT IT

looks like + feels like

DESIGN IS HOW IT

WORKS.

STEVE JOBS

SHORTIEDESIGNS®



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Web Designing Principles :

2. COMMUNICATION

People on the web tend to want information quickly, so it is important to communicate clearly, and make your information easy to read and digest. Some effective tactics to include in your web design include: organizing information using headlines and sub headlines, using bullet points instead of long windy sentences, and cutting the waffle.



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Web Designing Principles :

3. TYPEFACES

In general, Sans Serif fonts such as Arial and Verdana are easier to read online (Sans Serif fonts are contemporary looking fonts without decorative finishes). The ideal font size for reading easily online is 16px and stick to a maximum of 3 typefaces in a maximum of 3 point sizes to keep your design streamlined.

Serif

vs

Sans Serif.

TYPOGRAPHY

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Web Designing Principles :

4. COLOURS

A well thought out color palette can go a long way to enhance the user experience. Complementary colors create balance and harmony. Using contrasting colors for the text and background will make reading easier on the eye. Vibrant colors create emotion and should be used sparingly (e.g. for buttons and call to actions). Last but not least, white space/ negative space is very effective at giving your website a modern and uncluttered look.





Unit 1. Basic Web Technology

Web Designing Principles :

5. IMAGES

A picture can speak a thousand words, and choosing the right images for your website can help with brand positioning and connecting with your target audience. If you don't have high quality professional photos on hand, consider purchasing stock photos to lift the look of your website. Also consider using info graphics, videos and graphics as these can be much more effective at communicating than even the most well written piece of text.





Unit 1. Basic Web Technology

Web Designing Principles :

6. NAVIGATION

Navigation is about how easy it is for people to take action and move around your website. Some tactics for effective navigation include a logical page hierarchy, using bread crumbs, designing clickable buttons, and following the 'three click rule' which means users will be able to find the information they are looking for within three clicks.



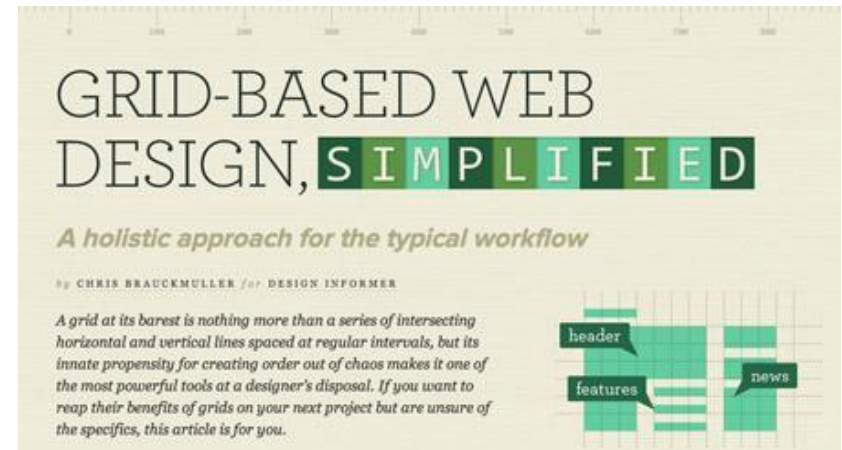


Unit 1. Basic Web Technology

Web Designing Principles :

7. GRID BASED LAYOUTS:

Placing content randomly on your web page can end up with a haphazard appearance that is messy. Grid based layouts arrange content into sections, columns and boxes that line up and feel balanced, which leads to a better looking website design.





Unit 1. Basic Web Technology

Web Designing Principles :

8. “F” PATTERN DESIGN:

Eye tracking studies have identified that people scan computer screens in an “F” pattern. Most of what people see is in the top and left of the screen and the right side of the screen is rarely seen. Rather than trying to force the viewer’s visual flow, effectively designed websites will work with a reader’s natural behavior and display information in order of importance (left to right, and top to bottom).



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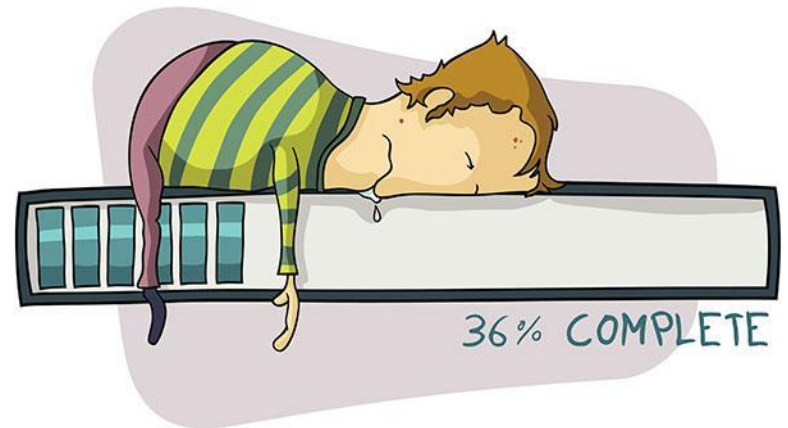


Unit 1. Basic Web Technology

Web Designing Principles :

9. LOAD TIME:

Everybody hates a website that takes ages to load. Tips to make page load times more effective include optimising image sizes (size and scale), combining code into a central CSS or JavaScript file (this reduces HTTP requests) and minify HTML, CSS, JavaScript (compressed to speed up their load time).





Unit 1. Basic Web Technology

Web Designing Principles :

10: MOBILE FRIENDLY:

It is now commonplace to access websites from multiple devices with multiple screen sizes, so it is important to consider if your website is mobile friendly. If your website is not mobile friendly, you can either rebuild it in a responsive layout (this means your website will adjust to different screen widths) or you can build a dedicated mobile site (a separate website optimized specifically for mobile users).





thank you

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